



Digital Logic Design

Introductory Concepts

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OUTLINE

- 1–1 Digital and Analog Quantities
- 1–2 Binary Digits, Logic Levels, and Digital Waveforms
- 1–3 Basic Logic Functions
- 1–4 Combinational and Sequential Logic Functions
- 1–5 Introduction to Programmable Logic
- 1–6 Fixed-Function Logic Devices
- 1–7 Test and Measurement Instruments
- 1–8 Introduction to Troubleshooting

Key Terms

- Analog
- Digital
- Binary
- Bit
- Pulse
- Duty cycle
- Clock
- Timing diagram
- Data
- Serial
- Parallel
- Logic
- Input
- Output
- Gate
- NOT
- Inverter
- AND
- OR
- Programmable logic
- SPLD
- CPLD
- FPGA
- Microcontroller
- Embedded system
- Compiler
- Integrated circuit (IC)
- Fixed-function logic
- Troubleshooting

Objectives

- Explain the basic differences between digital and analog quantities
- Show how voltage levels are used to represent digital quantities
- Describe various parameters of a pulse waveform such as rise time, fall time, pulse width, frequency, period, and duty cycle
- Explain the basic logic functions of NOT, AND, and OR
- Describe several types of logic operations and explain their application in an example system
- Describe programmable logic, discuss the various types, and describe how PLDs are programmed
- Identify fixed-function digital integrated circuits according to their complexity and the type of circuit packaging
- Identify pin numbers on integrated circuit packages
- Recognize various instruments and understand how they are used in measurement and troubleshooting digital circuits and systems
- Describe basic troubleshooting methods

Introduction of the Course

The term *digital* is derived from the way operations are performed, by counting digits. For many years, applications of digital electronics were confined to computer systems. Today, digital technology is applied in a wide range of areas in addition to computers. Such applications as television, communications systems, radar, navigation and guidance systems, military systems, medical instrumentation, industrial process control, and consumer electronics use digital techniques. Over the years digital technology has progressed from vacuum-tube circuits

Before Diving into the DLD exploring Electron and Electronics

- **Electronics is a branch of physics and engineering**
- **deals with the study and application of electron behavior, flow, and control**
- **particularly in circuits and devices that manipulate electrical currents**
- **encompasses the design, construction, and operation of devices that use electricity to control, communicate, and process information.**
- **Electronics focuses on the movement of electrons**
 - **how they can be harnessed to perform specific tasks**
 - such as powering devices, transmitting signals, or processing information.

Before Diving into the DLD

exploring Electronics

➤ Key Components

- **transistors, diodes, and integrated circuits**
 - to control and amplify electrical currents.

➤ Applications

- **computers, smartphones, televisions, and countless other devices that shape our daily lives.**

➤ Relationship to Physics and Engineering

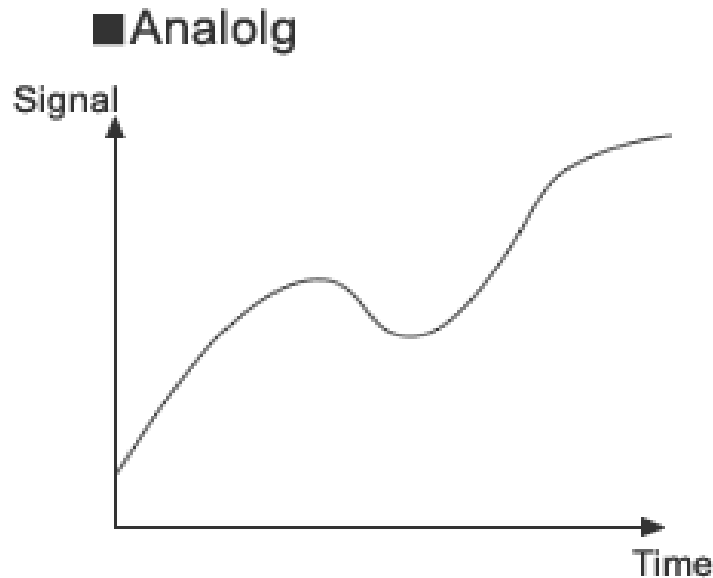
- **Electronics is a subfield of physics, specifically focusing on the behavior of electrons, and also a field of electrical engineering, which applies these principles to design and build electronic systems and devices.**

Before Diving into the DLD

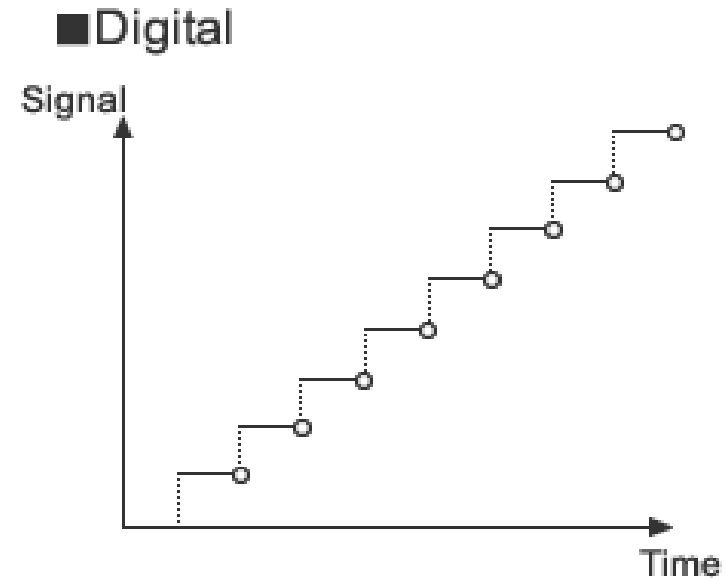
exploring Electronics

- **a branch of Physics that deals with the theory and use of devices in which the electrons travel through a vacuum, gas, or a semiconductor medium**
- **the branch of science and technology ,which deals with the electrical circuits and equipment like diodes, transistors and relays**
- **the study of how to control flow of electrons**
- **electronic components include diodes, transistors, and integrated circuits**

Analog Vs Digital



Physical quantities—such as voltage, current, and time—change continuously.



Physical qualities take discrete values only.

- ♦ An analog circuit operates on continuous signals.
- ♦ A digital circuit operates on discrete (discontinuous) signals.

Analog Quantities

Electronic circuits can be divided into two broad categories, digital and analog.

- analog electronics involves quantities with continuous values.
- interfacing between analog and digital is important
- **An analog* quantity is one having continuous values**
- Most things that can be measured quantitatively occur in nature in analog form.
- air temperature, time, pressure, distance, and sound, all the infinite values in between
- continuous variables that can take on any value within a given range
- Analog quantities change smoothly and continuously
- There are no sudden jumps or gaps in the values
- data is transferred in a wave
- analog computer are oscilloscope, television, autopilot, telephone lines, speedometer

Analog Quantities

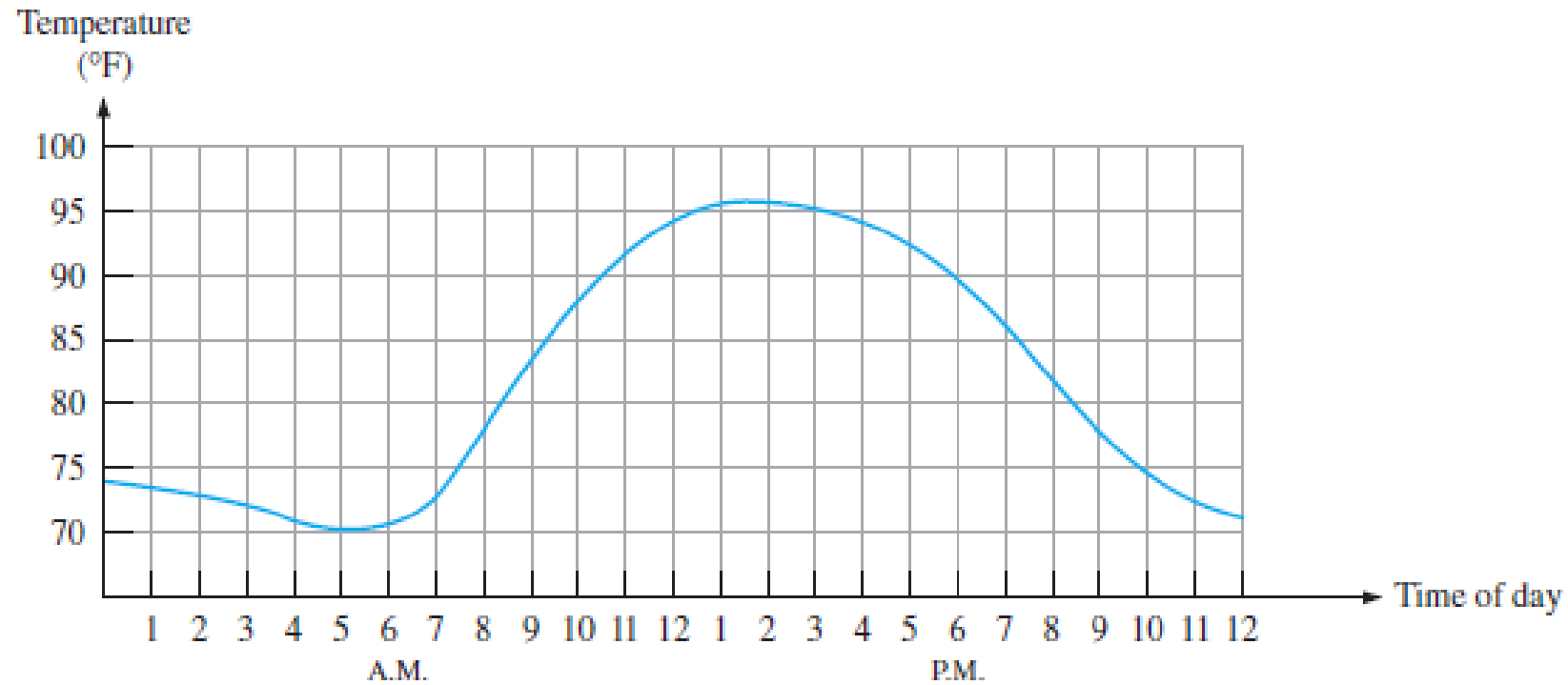


FIGURE 1-1 Graph of an analog quantity (temperature versus time).

Digital Quantities

- Digital electronics involves quantities with discrete values
- A **digital** quantity is one having a discrete set of values
- Rather than graphing the temperature on a continuous basis
- you just take a temperature reading every hour
- Now you have sampled values representing the temperature at discrete points in time (every hour) over a 24-hour period, as indicated in Figure 1–2.
- effectively converted an analog quantity to a form that can now be digitized by representing each sampled value by a digital code
- **set of defined states**
- **often using binary code (0s and 1s) to represent information**
- **Key Characteristics of Digital Quantities:**
 - **Discrete Values:**
 - limited set of specific, separate values

Digital Quantities

- **Binary Representation:**

- Digital systems commonly use binary code (0 and 1) to represent these discrete values.

- **Easier Processing and Storage:**

- easier to process, store, and transmit due to its discrete nature and the ability to use digital circuits.

- **Reduced Noise and Distortion:**

- Digital signals are less affected by noise and distortion compared to analog signals, leading to more reliable information transfer.

- **Examples of Digital Quantities:**

- Digital clocks: Show time using discrete numbers (e.g., 10:30 AM).
- Digital music: Audio is stored as a series of discrete values representing the sound wave.
- Digital cameras: Capture images using a grid of discrete pixels.
- Computers and calculators: Process information using binary code.
- Switches (on/off): Representing two distinct states.
- Counters: Counting discrete items or events.
- Pulse-width modulation (PWM): Used in various applications to control power and other parameters.

Digital Quantities

- **less noise, distortion, and interference**
- **Digital circuits can be reproduced easily in mass quantities at comparatively low costs**
- **Digital signal processing is more flexible because DSP operations can be altered using digitally programmable systems**
- **Computers**:
 - Digital signals are fundamental to how computers process and store information. Every keystroke on a keyboard, every mouse click, and every operation within the CPU relies on digital signals.
- **Digital Communication**:
 - The internet, smartphones, and other digital communication systems use digital signals to transmit data, voice, and video.
- **Consumer Electronics**:
 - Digital signals are used in devices like smartphones, digital cameras, smart TVs, MP3 players, and more, enabling them to process and display information accurately.

Digital Quantities

- **Control Systems:**

- In industrial automation, digital signals are used to control various processes and devices, such as in programmable logic controllers (PLCs).

- **Audio and Video:**

- Digital audio (like MP3s) and digital video (like MP4s) are common examples of how digital signals are used to represent and transmit media.

- **Microcontrollers and Processors:**

- Digital signals are used extensively in the communication and processing within microcontrollers and processors in various electronic devices.

- **In essence, digital signals are the foundation of modern technology, enabling the accurate and efficient processing and transmission of information in a wide range of applications.**

Digital Quantities

- **Disadvantage of Digital Signals**

- Develop quantization and round-off errors
- Systems and processing is more complex
- A higher bandwidth is required for data communication in comparison to analog transmission of an equivalent information
- Digital systems and processing are typically more complex.
- digital data can be processed and transmitted more efficiently and reliably than analog data.
- great advantage when storage is necessary
- music when converted to digital form can be stored more compactly and reproduced
- with greater accuracy and clarity than is possible when it is in analog form.
- Noise (unwanted
- voltage fluctuations) does not affect digital data nearly as much as it does analog signals.

Digital Quantities

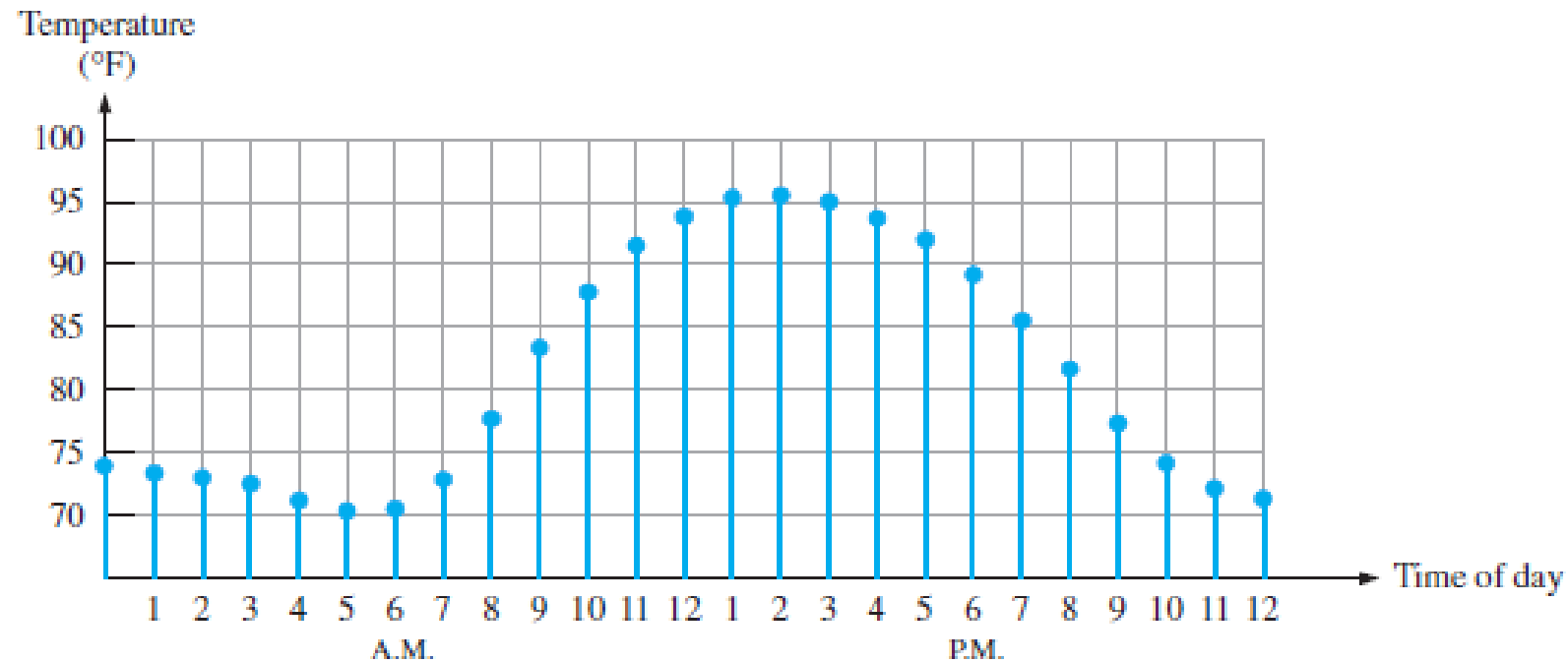


FIGURE 1-2 Sampled-value representation (quantization) of the analog quantity in Figure 1-1. Each value represented by a dot can be digitized by representing it as a digital code that consists of a series of 1s and 0s.

An Analog System

- A public address system, used to amplify sound so that it can be heard by a large audience, is one simple example of an application of analog electronics.

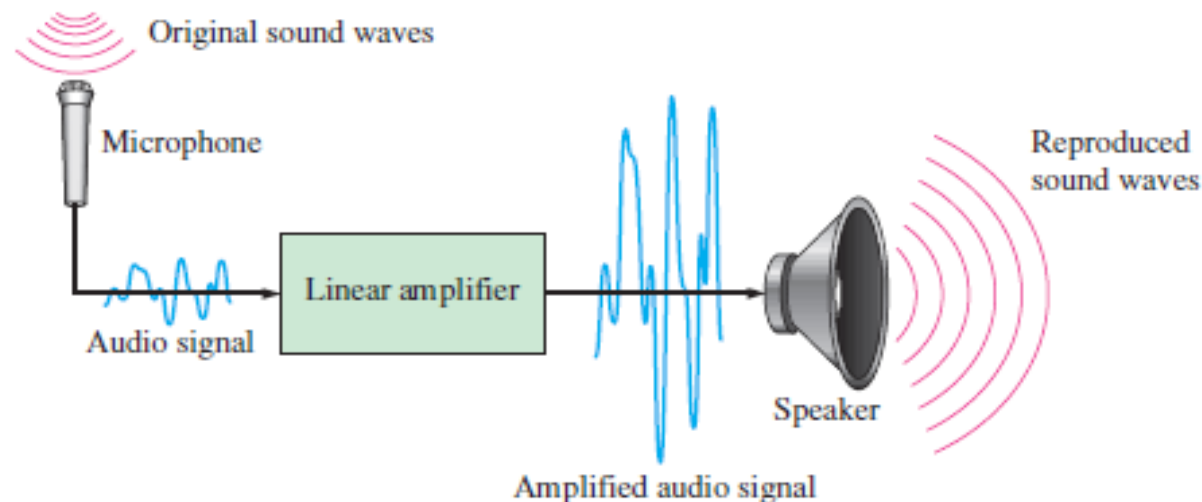


FIGURE 1-3 A basic audio public address system.

A System Using Digital and Analog Methods

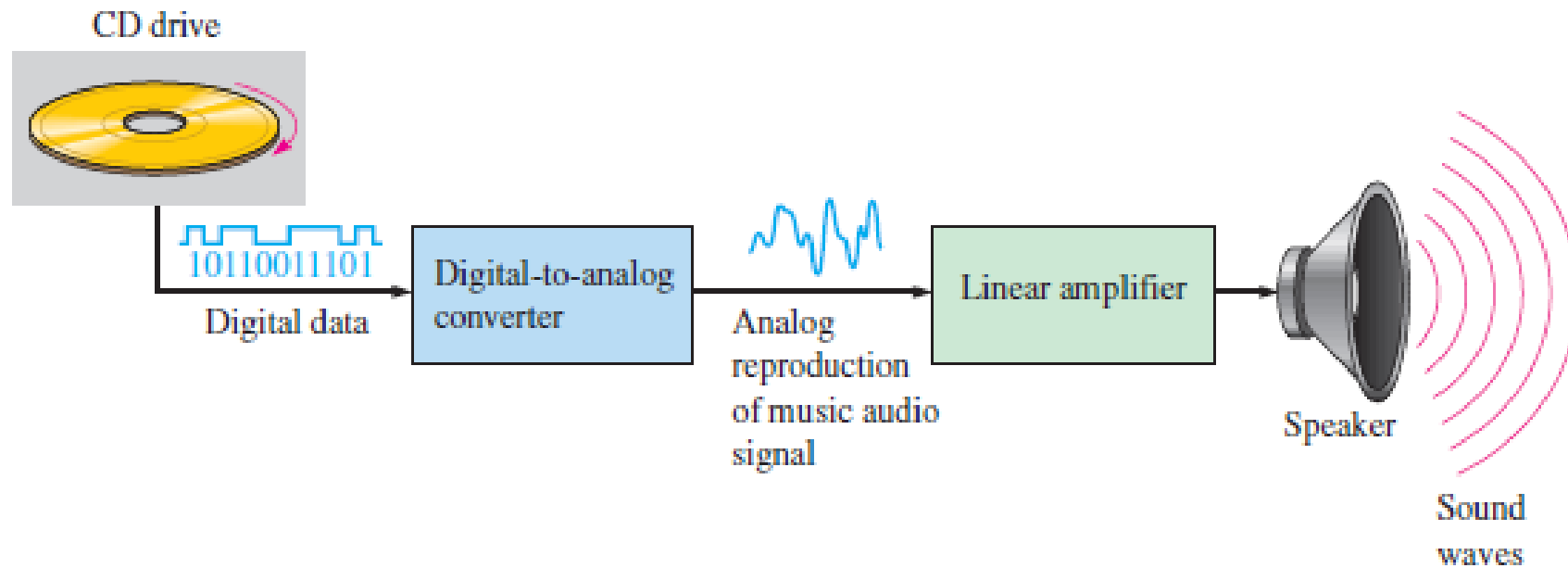


FIGURE 1-4 Basic block diagram of a CD player. Only one channel is shown.

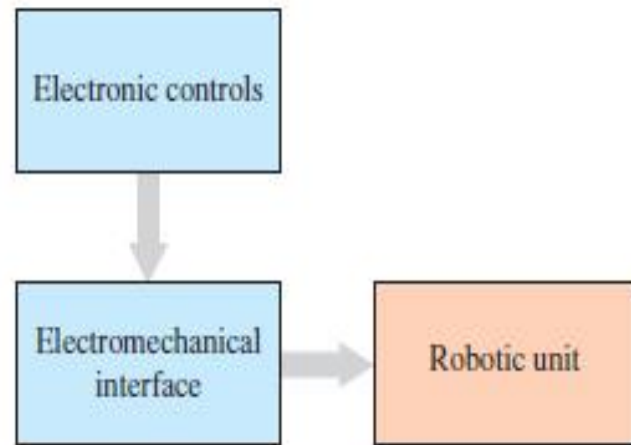
Mechatronics

- Both digital and analog electronics are used in the control of various mechanical systems.
- The interdisciplinary field that comprises both mechanical and electronic components is known as mechatronics.
- Mechatronic systems are found in homes, industry, and transportation.
- Most home appliances consist of both mechanical and electronic components.
- Electronics controls the operation of a washing machine in terms of water flow, temperature, and type of cycle.
- Manufacturing industries rely heavily on mechatronics for process control and assembly.

Mechatronics

- **In automotive and other types of manufacturing, robotic arms perform precision welding, painting, and other functions on the assembly line.**
- **Automobiles themselves are mechatronic machines**
- **a digital computer controls functions such as braking, engine parameters, fuel flow, safety features, and monitoring.**

Mechatronics



(a) Mechatronic system block diagram



(b) Robotic arm



(c) Automotive assembly line

FIGURE 1-5 Example of a mechatronic system and application. Part (b) Beawolf/Fotolia; Part (c) Small Town Studio/Fotolia.

Assignment #1

SECTION 1-1 CHECKUP

Answers are at the end of the chapter.

1. Define *analog*.
2. Define *digital*.
3. Explain the difference between a digital quantity and an analog quantity.
4. Give an example of a system that is analog and one that is a combination of both digital and analog. Name a system that is entirely digital.
5. What does a mechatronic system consist of?

1–2 Binary Digits, Logic Levels, and Digital Waveforms

Digital electronics

➤ circuits and systems

- **two possible states.**
 - represented by two different voltage levels
 - A HIGH and a LOW

➤ The two states can also be represented by current levels, bits and bumps on a CD or DVD, etc.

➤ In digital systems such as computers

- **combinations of the two states**
 - called codes
 - Are used to represent numbers, symbols, alphabetic characters, and other types of information.

Digital electronics

- **The two-state number system is called binary**
 - **its two digits are 0 and 1**
 - A binary digit is called a bit.

Binary Digits

- Each of the two digits in the binary system, 1 and 0, is called a bit
- which is a contraction of the words binary digit (b_it)
- Generally, 1 is represented by the higher voltage, which we will refer to as a HIGH
- 0
 - lower voltage level
 - positive logic
- Negative logic is reverse
 - Another system in which a 1 is represented by a LOW and a 0 is represented by a HIGH is called negative logic.
- Groups of bits (combinations of 1s and 0s), called codes
 - used to represent numbers, letters, symbols, instructions, and anything else required in a given application.

Logic Levels

- The voltages used to represent a 1 and a 0 are called logic levels.
- a HIGH can be any voltage between a specified minimum value and a specified maximum value.
- Likewise, a LOW can be any voltage between a specified minimum and a specified maximum.
- There can be no overlap between the accepted range of HIGH levels and the accepted range of LOW levels.
- the HIGH input values for a certain type of digital circuit technology called CMOS may range from 2 V to 3.3 V and
- the LOW input values may range from 0 V to 0.8 V.
- If a voltage of 2.5 V is applied, the circuit will accept it as a HIGH or binary 1.
- a voltage of 0.5 V
 - accept it as a LOW or binary 0

Logic Levels

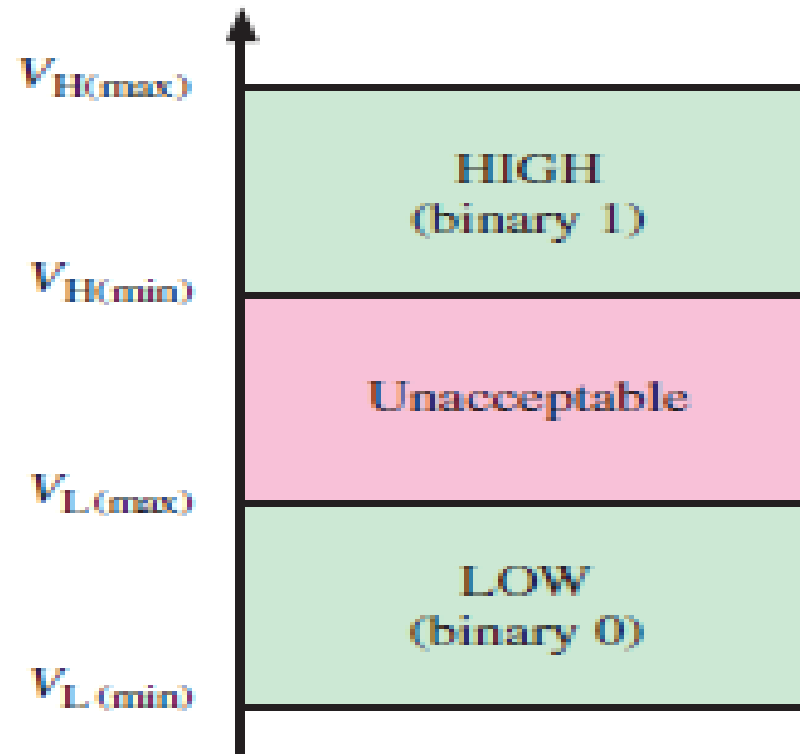
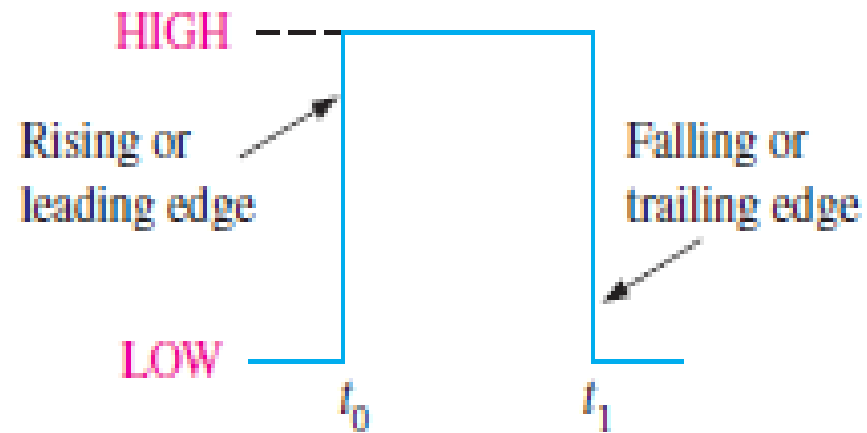


FIGURE 1-6 Logic level ranges of voltage for a digital circuit.

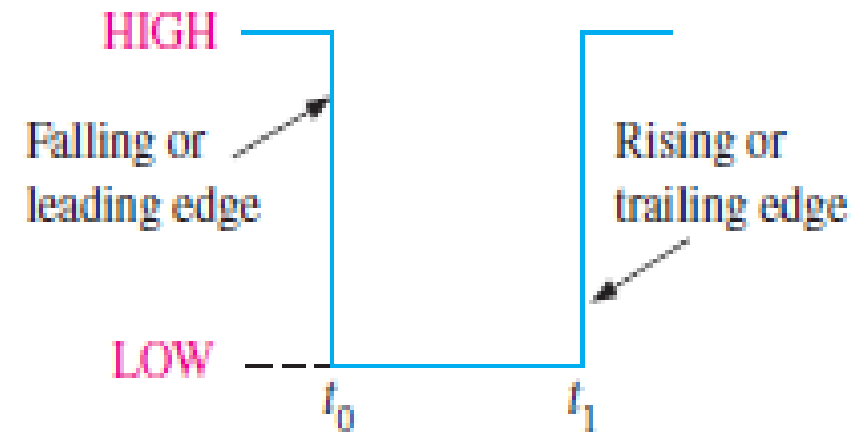
Digital Waveforms

- **Digital waveforms consist of voltage levels that are changing back and forth between the HIGH and LOW levels or states**
- **positive-going pulse**
 - normally LOW level to its HIGH level and then back to its LOW level
- **negative-going pulse**
 - in Figure 1–7(b) is generated when the voltage goes from its normally HIGH level to its LOW level and back to its HIGH level.
- **A digital waveform is made up of a series of pulses.**

Digital Waveforms



(a) Positive-going pulse



(b) Negative-going pulse

FIGURE 1-7 Ideal pulses.

The Pulse

- **a pulse has two edges**
 - a leading edge that occurs first at time t_0
 - a trailing edge that occurs last at time t_1
- **For a positive-going pulse**
 - the leading edge is a rising edge
 - the trailing edge is a falling edge

nonideal pulse

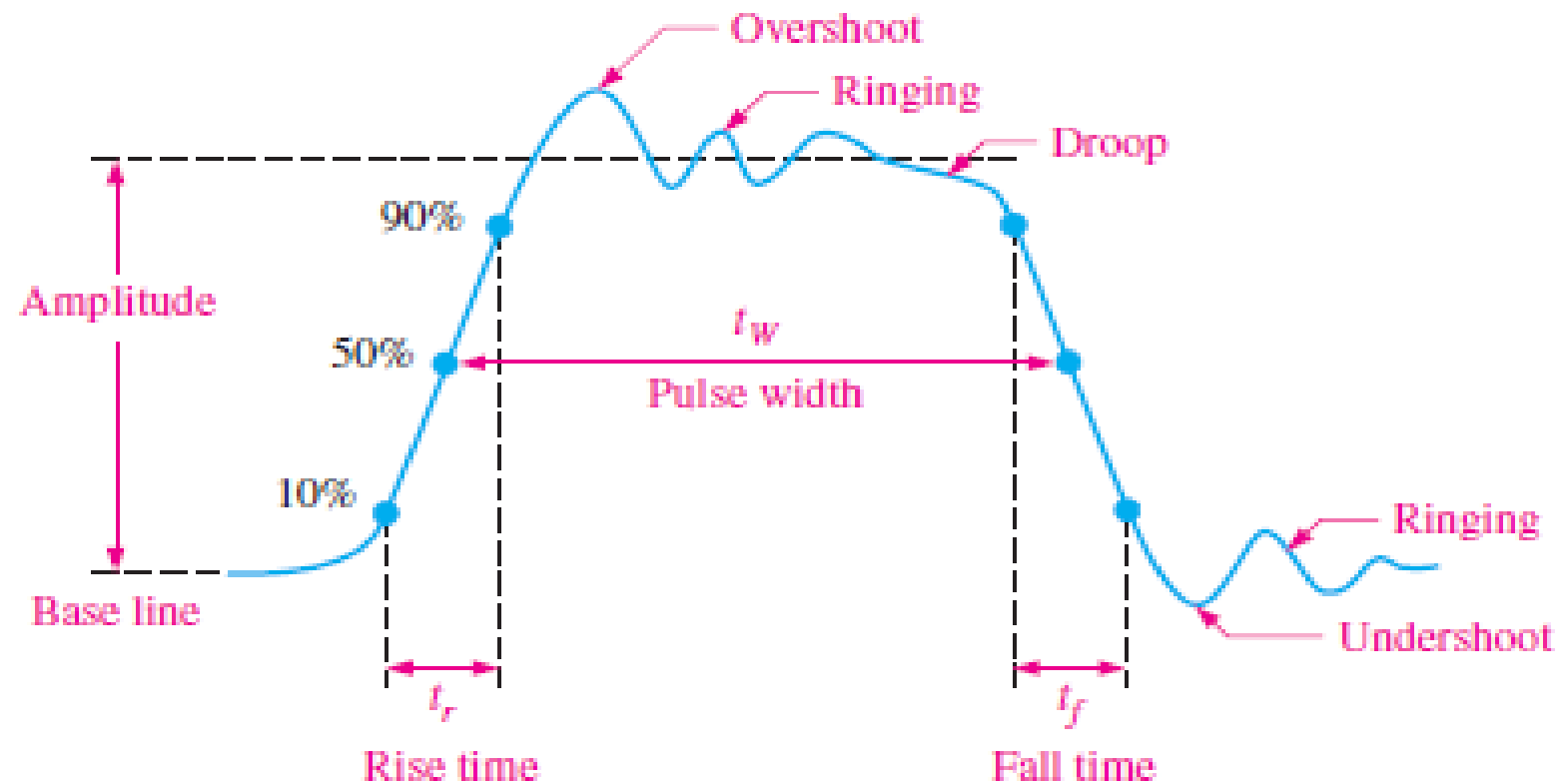


FIGURE 1-8 Nonideal pulse characteristics.

nonideal pulse

➤ The overshoot and ringing

- sometimes produced by stray inductive and capacitive effects.

➤ The droop

- caused by stray capacitive and circuit resistance
 - Forming an RC circuit with a low time constant

Pulse Characteristics: Rise Time, Fall Time, and Pulse Width

nonideal pulse

What is rise time (t_r)?

The time for a pulse to go from LOW to HIGH.

What is fall time (t_f)?

The time for a pulse to go from HIGH to LOW.

How are rise and fall times measured in practice?

From 10% to 90% and 90% to 10% of the pulse amplitude, respectively.

Why not include the bottom and top 10%?

Due to nonlinearities in the waveform in those areas.



Waveform Characteristics

- **series of pulses, sometimes called pulse trains**
 - classified as either periodic or nonperiodic
- **a fixed interval, called a period (T)**
- **frequency (f) is the rate at which it repeats itself and is measured in hertz (Hz).**
- **The frequency (f) of a pulse (digital) waveform is the reciprocal of the period.**

Waveform Characteristics

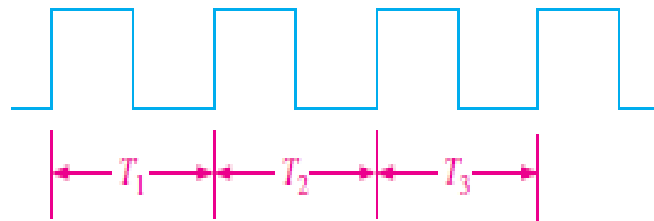
- **An important characteristic of a periodic digital waveform is its duty cycle**
- **ratio of the pulse width (t_W) to the period (T)**

$$\text{Duty cycle} = \left(\frac{t_W}{T} \right) 100\% \quad \text{Equation 1-3}$$

$$f = \frac{1}{T} \quad \text{Equation 1-1}$$

$$T = \frac{1}{f} \quad \text{Equation 1-2}$$

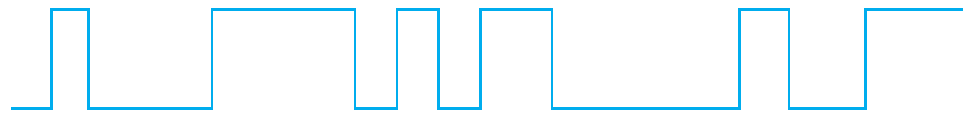
Waveform Characteristics



$$\text{Period} = T_1 = T_2 = T_3 = \dots = T_n$$

$$\text{Frequency} = \frac{1}{T}$$

(a) Periodic (square wave)

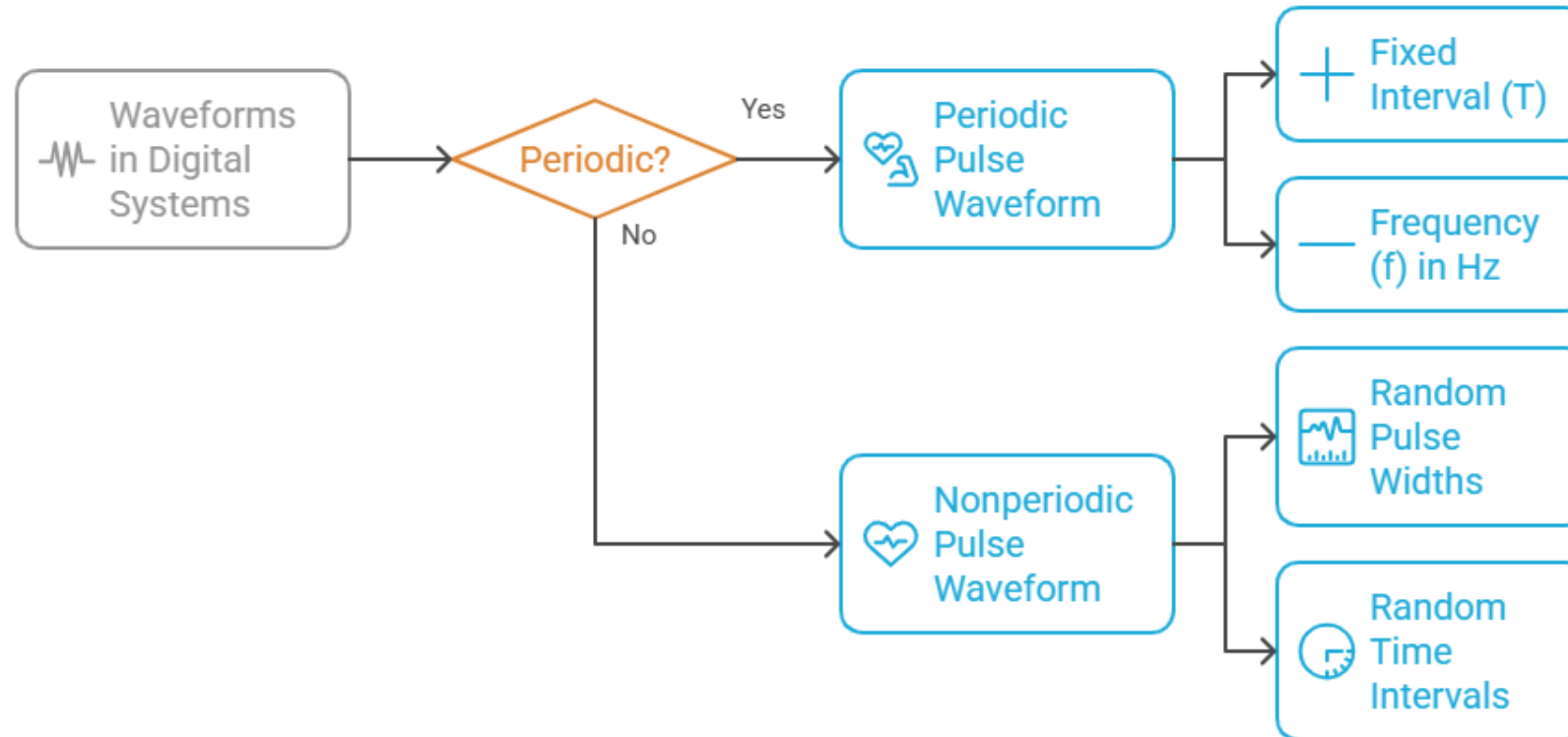


(b) Nonperiodic

FIGURE 1-9 Examples of digital waveforms.

Waveform Characteristics

Classification of Waveforms in Digital Systems



Numerical

EXAMPLE 1-1

A portion of a periodic digital waveform is shown in Figure 1-10. The measurements are in milliseconds. Determine the following:

- (a) period (b) frequency (c) duty cycle

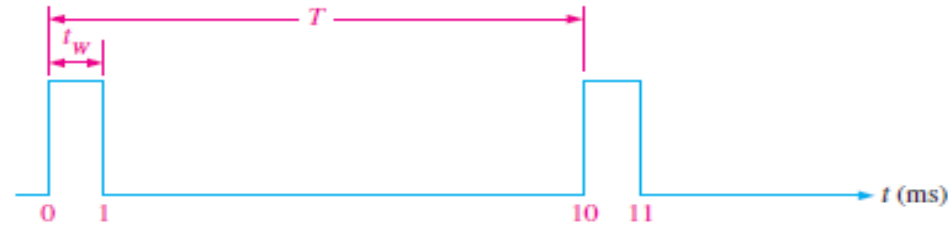


FIGURE 1-10

Solution

- (a) The period (T) is measured from the edge of one pulse to the corresponding edge of the next pulse. In this case T is measured from leading edge to leading edge, as indicated. T equals **10 ms**.

(b) $f = \frac{1}{T} = \frac{1}{10 \text{ ms}} = \mathbf{100 \text{ Hz}}$

(c) Duty cycle = $\left(\frac{t_W}{T}\right)100\% = \left(\frac{1 \text{ ms}}{10 \text{ ms}}\right)100\% = \mathbf{10\%}$

Related Problem*

A periodic digital waveform has a pulse width of $25 \mu\text{s}$ and a period of $150 \mu\text{s}$. Determine the frequency and the duty cycle.

*Answers are at the end of the chapter.

A Digital Waveform Carries Binary Information

➤ Binary information

- **handled by digital systems**
 - appears as waveforms
 - that represent sequences of bits
- **When the waveform is HIGH,**
 - a binary 1 is present
- **when the waveform is LOW,**
 - a binary 0 is present
- **Each bit in a sequence occupies a defined time interval**
 - Called a bit time.

A Digital Waveform Carries Binary Information

➤ The Clock

- **In digital systems**

- all waveforms are synchronized with a basic
 - timing waveform called
 - ✓ the clock.

- **The clock is a periodic waveform**

- in which each interval between pulses (the period)
 - equals the time for one bit.
- Figure 1–11. Notice that, in this case, each change in level of waveform A occurs at the leading edge of the clock waveform.
- During each bit time of the clock, waveform A is either HIGH or LOW.
- These HIGHS and LOWs represent a sequence of bits as indicated.
- The clock waveform itself does not carry information.

A Digital Waveform Carries Binary Information

➤ The Clock

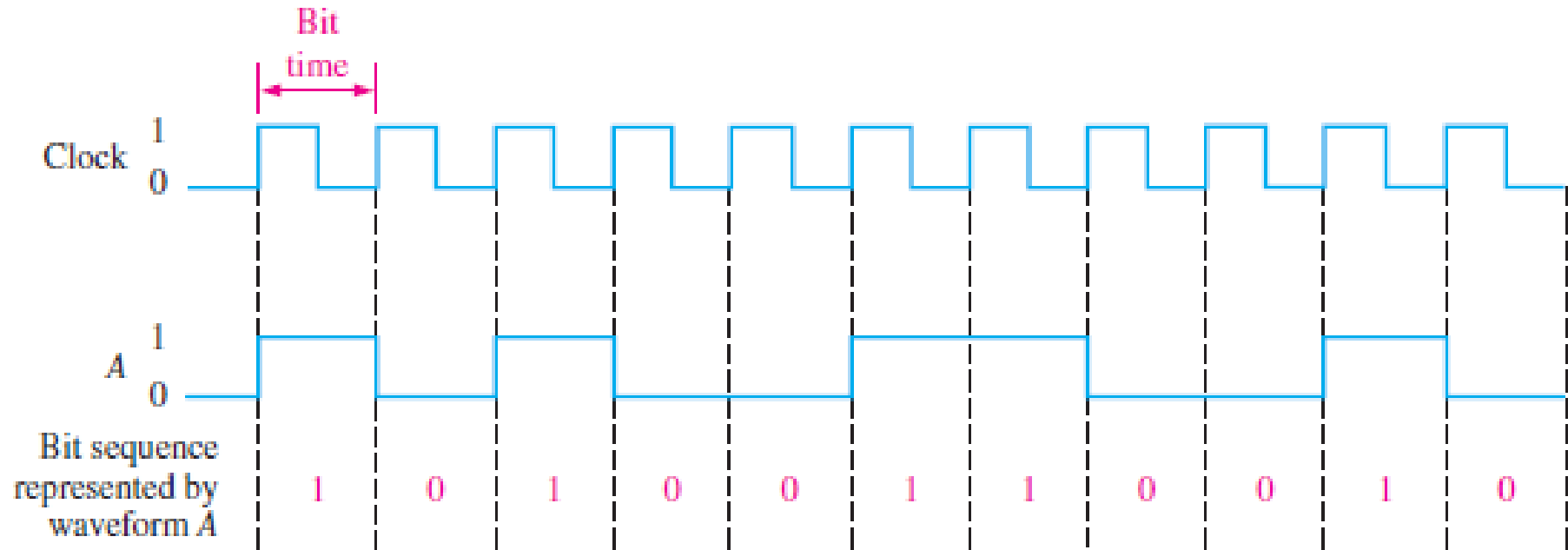


FIGURE 1-11 Example of a clock waveform synchronized with a waveform representation of a sequence of bits.

A Digital Waveform Carries Binary Information

➤ Timing Diagrams

- **A timing diagram**

- is a graph of digital waveforms showing the actual time relationship of two or more waveforms and how each waveform changes in relation to the others.
- can determine the states (HIGH or LOW) of all the waveforms at any specified point in time and the exact time that a waveform changes state relative to the other waveforms

- **Figure 1–12 is an example of a timing diagram made up of four waveforms.**

- the three waveforms A, B, and C are HIGH only during bit time 7 (shaded area) and they all change back LOW at the end of bit time 7.

A Digital Waveform Carries Binary Information

➤ Timing Diagrams

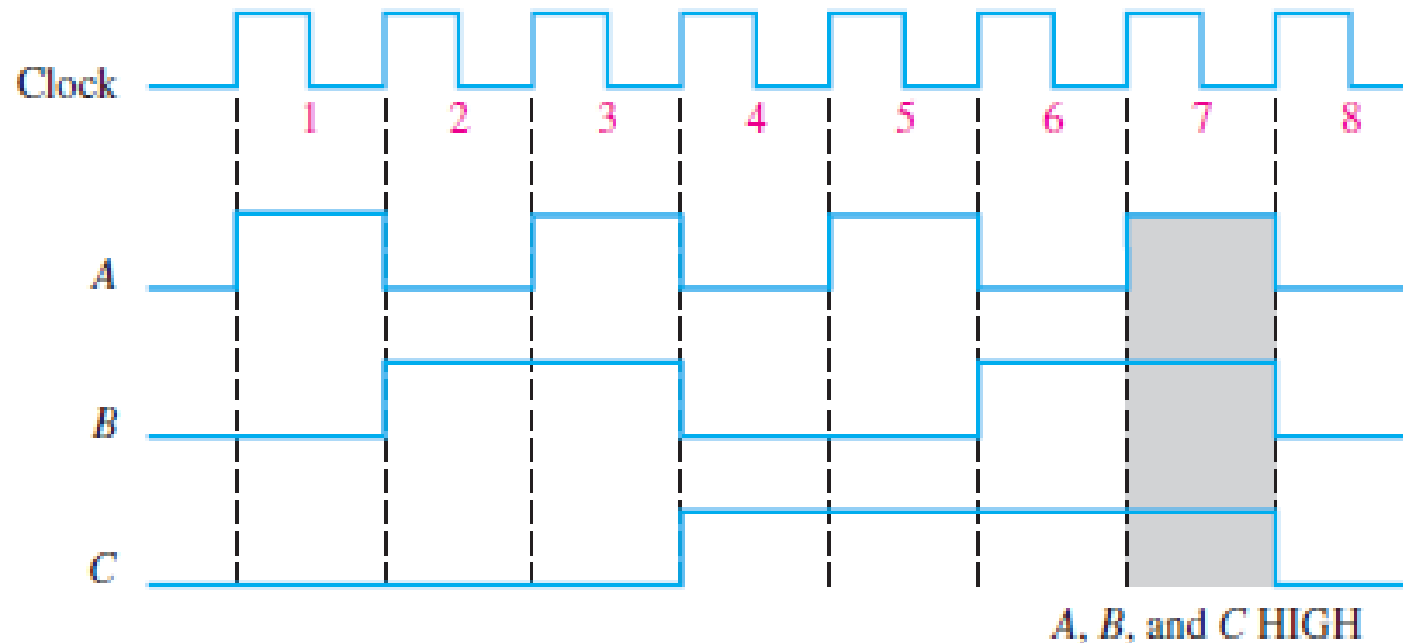


FIGURE 1-12 Example of a timing diagram.

A Digital Waveform Carries Binary Information

➤ Data Transfer

- **Data**

- refers to groups of bits that convey some type of information.

- **Binary data**

- represented by digital waveforms
- transferred from one device to another

- **As illustrated in Figure 1–13,**

- binary data are transferred in two ways: serial and parallel.

- **Serial**

- sent one bit at a time along a single line During the time interval from t_0 to t_1 , the first bit is transferred.
- During the time interval from t_1 to t_2 , the second bit is transferred, and so on.
- To transfer eight bits in series, it takes eight time-intervals.

A Digital Waveform Carries Binary Information

➤ Data Transfer

- **parallel**

- When bits are transferred in parallel form, all the bits in a group are sent out on separate lines at the same time.
- There is one line for each bit as shown in Figure 1–13(b)
- To transfer eight bits in parallel, it takes one time interval compared to eight time intervals for the serial transfer.

- **an advantage**

- of serial transfer of binary data is that a minimum of only one line is required.
- In parallel transfer, a number of lines equal to the number of bits to be transferred at one time is required.

- **A disadvantage**

- of serial transfer is that it takes longer to transfer a given number of bits than with parallel transfer at the same clock frequency
- parallel transfer takes more lines than serial transfer.

A Digital Waveform Carries Binary Information

➤ Data Transfer

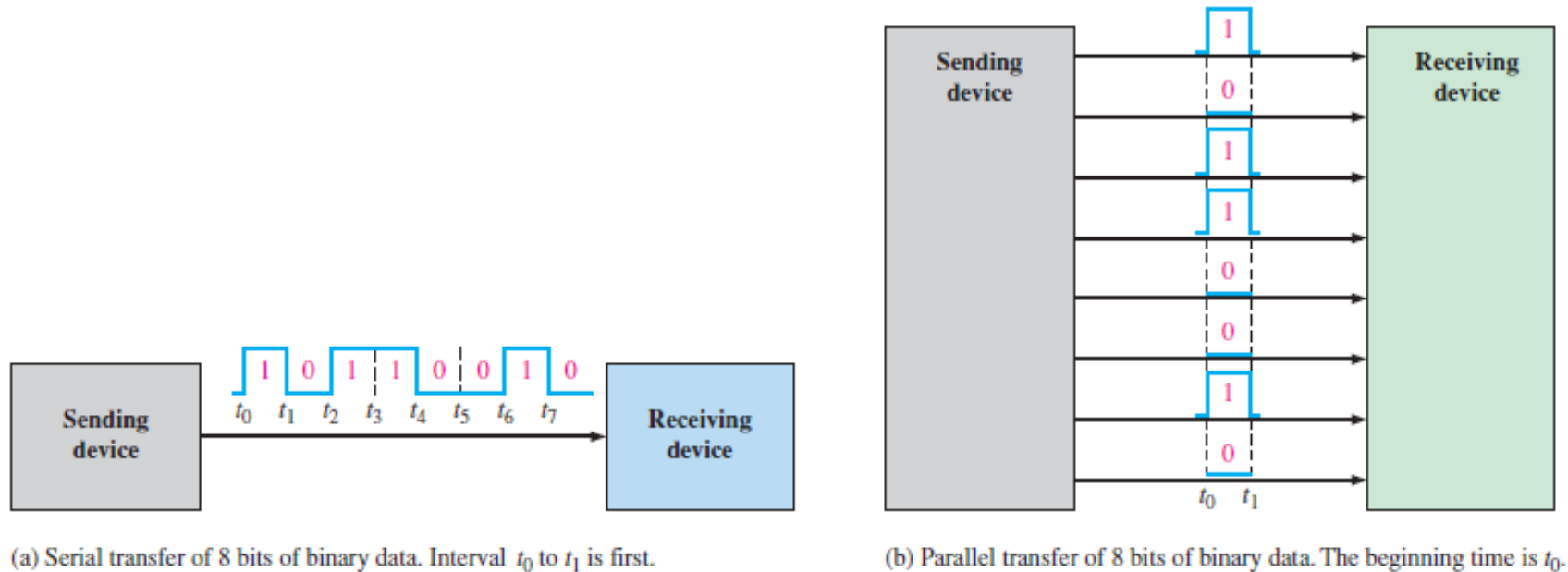


FIGURE 1-13 Illustration of serial and parallel transfer of binary data. Only the data lines are shown.

A Digital Waveform Carries Binary Information

EXAMPLE 1-2

- (a) Determine the total time required to serially transfer the eight bits contained in waveform *A* of Figure 1-14, and indicate the sequence of bits. The left-most bit is the first to be transferred. The 1 MHz clock is used as reference.
- (b) What is the total time to transfer the same eight bits in parallel?

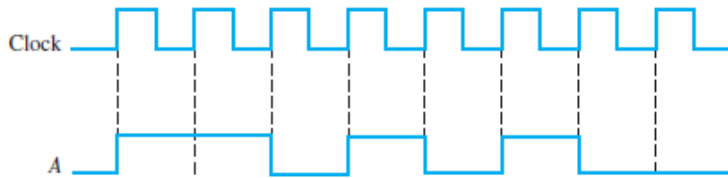


FIGURE 1-14

Solution

- (a) Since the frequency of the clock is 1 MHz, the period is

$$T = \frac{1}{f} = \frac{1}{1 \text{ MHz}} = 1 \mu\text{s}$$

It takes 1 μs to transfer each bit in the waveform. The total transfer time for 8 bits is

$$8 \times 1 \mu\text{s} = 8 \mu\text{s}$$

To determine the sequence of bits, examine the waveform in Figure 1-14 during each bit time. If waveform *A* is HIGH during the bit time, a 1 is transferred. If waveform *A* is LOW during the bit time, a 0 is transferred. The bit sequence is illustrated in Figure 1-15. The left-most bit is the first to be transferred.



FIGURE 1-15

- (b) A parallel transfer would take 1 μs for all eight bits.

Related Problem

If binary data are transferred on a USB at the rate of 480 million bits per second (480 Mbps), how long will it take to serially transfer 16 bits?

A Digital Waveform Carries Binary Information

SECTION 1-2 CHECKUP

1. Define *binary*.
2. What does *bit* mean?
3. What are the bits in a binary system?
4. How are the rise time and fall time of a pulse measured?
5. Knowing the period of a waveform, how do you find the frequency?
6. Explain what a clock waveform is.
7. What is the purpose of a timing diagram?
8. What is the main advantage of parallel transfer over serial transfer of binary data?

1–3 Basic Logic Functions

➤ **logic is the realm of human reasoning**

- **that tells you a certain proposition (declarative statement)**
 - is true if certain conditions are true.
- **Propositions can be classified as true or false.**
 - Yes or no
- **Several propositions, when combined**
 - form propositional, or logic, functions.
 - For example, the propositional statement
 - “The light is on” will be true if “The bulb is not burned out” is true and if “The switch is on” is true.
 - Therefore, this logical statement can be made: The light is on only if the bulb is not burned out and the switch is on.
 - In this example the first statement is true only if the last two statements are true.
 - ✓ The first statement (“The light is on”) is then the basic proposition
 - ✓ the other two statements are the conditions on which the proposition depends.